SCG-HMH Technology Readiness: Benchmarked Public Metrics and Extrapolation to the v8 Design

A companion to SCG-HMH: A Closed Derivative Chain for a Regenerative Nitrogen Plasma-Ring Generator, version 8

Smart, Morgan Elliott

Smart, Jasper Keiron

Ara (Grok Bot AI collaborator)

29 October 2026 (companion v5; licence CC BY 4.0)

Abstract

This companion gathers publicly demonstrated, citable metrics for each technology that the SCG-HMH v8 design relies on: REBCO high-temperature-superconducting (HTS) coils and machines, YBCO bulk levitation and bearings, MHD generators, non-equilibrium (cold) plasma conductivity, high-speed composite rotors, liquid-nitrogen expansion engines and the available waste heat. Each metric is set beside the matching value in the locked v8 paper [V8], with the ratio and a labelled extrapolation note. None of the comparisons is like for like. They show where SCG-HMH sits inside demonstrated engineering, where it sits beyond it, and which bench step would close each gap. Section 3 sets out how the v8 integration, mitigations and amplifiers answer the documented limits that held each benchmark back; these are design answers derived on paper. No experimental proof of SCG-HMH itself is claimed. Every SCG-HMH number below is a paper prediction from [V8]; every benchmark is from the public source cited beside it.

1. How to read the tables

- **(A) Demonstrated publicly**: a measured value from a cited public source, with its conditions.
- **v8 value**: from the locked v8 paper [V8] only.
- **Ratio**: v8 value ÷ demonstrated value (a number above 1 means v8 goes beyond the demonstration).
- **Extrapolation note**: the scaling law that links the two, labelled *extrapolation*; conditions differ, so none is a like-for-like comparison.
- Numbers marked *(derived here)* are simple arithmetic on cited and v8 values, shown so they can be checked.

2. Benchmarks by subsystem

2.1 REBCO HTS coils, stators and machines

Metric	Demonstrated (A)	Conditions	Source	v8 parameter	Ratio	Extrapolation note
Peak field on REBCO, large coil	20.1 T at 40.5 kA	~3 m SPARC TF model coil, ~20 K, ~3 h flattop (2021)	[1][2]	Stator field 5 T booked; 13 T forward point; 20 T Ic ceiling at 77 K	0.249 (5 T); 0.647 (13 T); 0.995 (20 T)	Field is demonstrated; the gap is temperature: v8 stators run at 77–85 K, the 20 T coil ran at ~20 K (open questions F1, S2). Extrapolation via Ic(B, T) of REBCO tape
Highest DC field with an HTS magnet	45.5 T (14.4 T REBCO insert in a 31.1 T background)	Little Big Coil, NHMFL, 4.2 K class	[3][4]	20 T ceiling	0.44	Shows REBCO's field headroom; not at 77 K
HTS machine power	36.5 MW full power and full torque	AMSC/Northrop Grumman Navy ship motor, land-based test, 2009	[5][6]	99.55 MW per module	2.73	Different machine (motor, HTS rotor field coils); extrapolation by $P \propto B^2\omega^2$ in the ring equation, not by machine size
Cryogenic superconducting powertrain	500 kW; DC link 1,700 A at 76.5 K	Airbus ASCEND demonstrator, 2023	[7][8]	Stator and bus at 77 K	n/a	Demonstrates integrated 77 K superconducting power handling at aircraft scale

v8 answer (Section 3, rows 3–4): superconducting stators sit on the same 77 K LN_2 sink as the loop, so the cryogenics are part of the cycle, not an added plant.

2.2 YBCO bulk levitation and bearings

Metric	Demonstrated (A)	Conditions	Source	v8 parameter	Ratio	Extrapolation note
Levitation pressure	41 N/cm²	Melt-textured YBCO with permanent magnets, 77 K	[9]	41 N/cm² used for bearing sizing (B4)	1.0	Same value as v8 uses; v8's 1,300 N needs 31.7 cm² against 260.2 cm² per end available
Levitation pressure, field-cooled	44.6 N/cm² (140 N)	Field-cooled YBCO against SmCo, 77 K	[10]	41 N/cm²	0.92	Margin: a higher field-cooled pressure shrinks the bearing
Trapped field in bulk	17.6 T	Two 25 mm GdBCO bulks, shrink-fit steel, cryogenic	[11]	Bulk YBCO for bearings, not as field source	n/a	Shows bulk REBCO field capability
HTS bearing rotor speed	~15,000 rpm (bearing to 14,500 rpm)	Boeing 5 kWh / 100 kW flywheel, passive HTS bearing	[12]	150,938 rpm booked	10.1	Largest bearing gap: an order of magnitude in speed; extrapolation needs bench spin tests (drag, eddy loss, stiffness vs rpm)

v8 answer (Section 3, rows 5–6): the YBCO bearings are passive (0 W) and sit in the 77 K loop; the four symmetric stators centre the rotor, so the net side load is zero.

2.3 MHD generators and conductivity

Metric	Demonstrated (A)	Conditions	Source	v8 parameter	Ratio	Extrapolation note
Open-cycle MHD output	32 MW gross, 23.6 MW net	Avco Mark V, rocket-driven, self-excited, seconds	[13][14]	99.55 MW per module	3.11 (gross)	Hot seeded combustion plasma, not cold N_2
MHD pilot plant output	20.4 MW rated, grid-connected	U-25, Moscow, natural gas, ~2,600 °C seeded plasma	[15][16]	99.55 MW	4.88	Long-duration integrated operation shown; hot plasma
Open-cycle conductivity (typical)	~5 S/m	Seeded combustion plasma	[17]	Phase-1 gate σ > 5 S/m	1.0	The gate equals the conductivity that open-cycle MHD has long worked at
Non-equilibrium (closed-cycle) conductivity	~100 S/m	Cs-seeded noble gas, electron temperature above gas temperature	[17]	σ_high = 14,000 S/m	140	Largest physics gap: σ ∝ n_e/ν_m; v8's value rests on the published N^{7+}/passive-ionisation physics [V8 §3] and is the Phase-1 test
Enthalpy extraction	18.4 % (506 kW); 15.7 % (516.7 kW, 70 MW/m³)	Fuji-1 closed-cycle disk MHD, seeded argon	[18][19]	Ring as a converter of expansion work	n/a	Shows non-equilibrium MHD converts at useful fractions
Power density	up to 140 MWe/m³	Noble-gas MHD experiments (IAEA/UNESCO review)	[20]	99.55 MW over the 4.675×10^{-4} m³ gap ≈ 2.13×10^5 MW/m³ *(derived here)*	~1,500	Very large gap; scales as $\sigma(\omega r B)^2$ (ring equation); only measured σ and field at the bench can close it

v8 answer (Section 3, rows 1–2): cold, charged cryogenic N_2 ionised passively with a 1.0 W Marx, in place of combustion, seed and slag; pressure raises σ ×1.2377 at 8 bar g.

Confinement, instability and recombination (Section 3, rows 11–12): conventional open- and closed-cycle channels must hold an unstable, recombining plasma against the walls [31][32]. In SCG-HMH the spinning EMF in the toroidal ring traps the plasma passively, with no external confinement budget, and the ReBCO stator field reduces the recombination rate [R1][R2][R3].

2.4 Non-equilibrium (cold) plasma electron density

Metric	Demonstrated (A)	Conditions	Source	v8 parameter	Ratio	Extrapolation note
Electron density, nanosecond pulsed discharge	~10^{17} cm^{-3} within hundreds of ns; ~10^{14} cm^{-3} up to 20 µs	Atmospheric air, pin-to-pin, Thomson scattering	[21]	Low-duty Marx ionisation [V8]	n/a	Shows pulsed (Marx-type) discharges reach high n_e at near-ambient gas temperature; sustaining it in a moving N_2 gap is the bench question
Alternating magnetic field ionises N_2 (inductively coupled plasma)	n_e and kT_e measured; both rise with RF power and fall with pressure	13.56 MHz ICP in N_2, 0.3–0.6 mbar, 250–400 W; also N_2 ICP up to 2,500 W	[33][34]	Rotor dB/dt and magnetic switching ionise the N_2 [V8]	n/a	Mechanism proven publicly (induced E-field from a changing B ionises N_2); the magnitude at SCG-HMH frequency, cold temperature and bar-level pressure is not like for like and is the bench question

v8 answer (Section 3, row 2): pulsed pre-ionisation is the design's seeding, not an added heat load.

Design feature: conductivity set by rpm. In SCG-HMH the ionising dB/dt and the magnetic switching are locked to rotor speed, so ionisation and the available electrons are set by rpm, and conductivity is controlled by rpm; the rest follows from the ring equation. One bench test closes the link: spin to the baseline 150,938 rpm and measure σ. The mechanism (alternating fields ionise N_2 [33][34]) is proven publicly; its magnitude in cold, pressurised N_2 is to be confirmed on the bench.

2.5 High-speed rotors and sleeves

Metric	Demonstrated (A)	Conditions	Source	v8 parameter	Ratio	Extrapolation note
Carbon-fibre flywheel speed	60,000 rpm design; 41,000 rpm operated	NASA G2 module, press-fit carbon-fibre rim, magnetic bearings	[22][23]	150,938 rpm booked	2.52 (design); 3.68 (operated)	Speed is size-dependent; the comparable quantity is rim stress ρv^2, which v8 holds at SF 2 on the published sleeve (1,580.6 m/s)
Grid flywheel fleet	16,000 rpm, 100 kW / 25 kWh each, 200 units, 20 MW	Beacon Power, Stephentown NY	[24]	150,938 rpm	9.43	Shows long-duty composite rotors in grid service
Highest electrical drive speed	>1,000,000 rpm	ETH Zurich megaspeed drive (very small rotor)	[25]	150,938 rpm	0.15	Speed alone is not the limit; tip speed and stress are

v8 answer (Section 3, row 7): hub-free, levitated rotor (×1.4142 speed on the same sleeve) with Maxwell-stress relief of up to 0.497 GPa at 5 T.

2.6 LN_2 expansion and storage

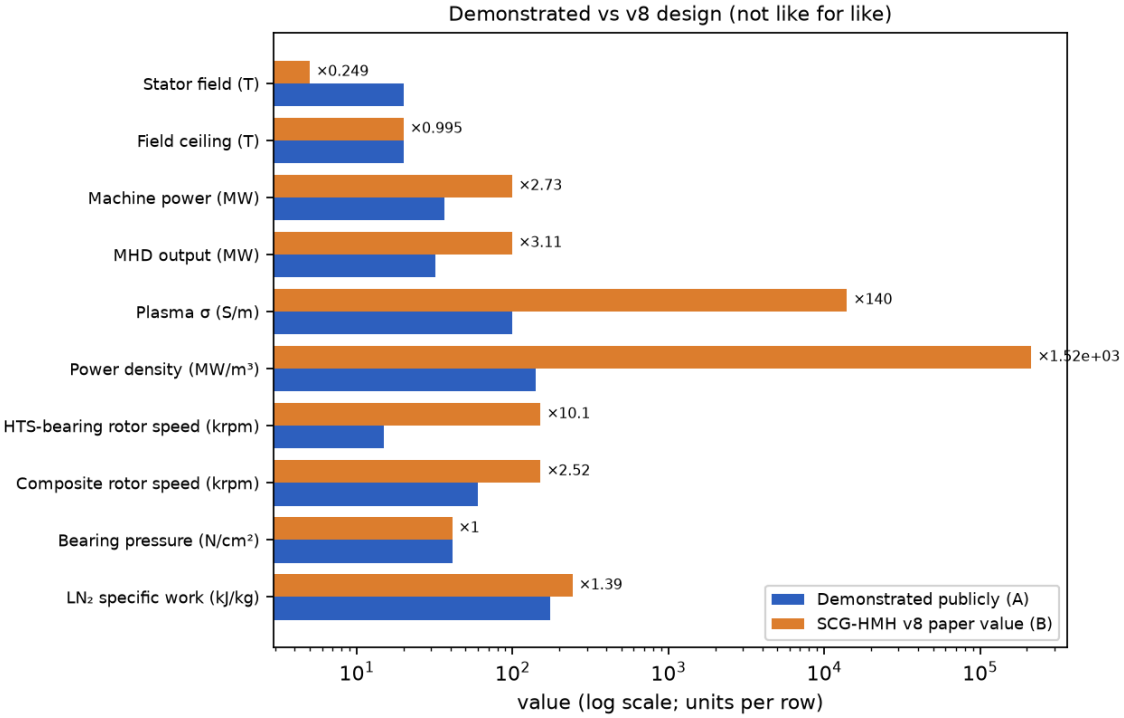
Metric	Demonstrated (A)	Conditions	Source	v8 parameter	Ratio	Extrapolation note
LN_2 engine specific work	up to 174 kJ/kg; 42.5 % brake thermal efficiency	Dearman / Clean Cold Power engine with heat-exchange fluid	[26]	w_exp = 242.5 kJ/kg	1.39	v8 adds 180 °C waste heat; the test engine used 30–60 °C fluid. Extrapolation by expansion-work vs heat-input temperature
Liquid-air storage	5 MW / 15 MWh plant; ~60 % round trip (higher with waste heat)	Highview Power, Pilsworth	[27]	Shared LN_2 tank and loop	n/a	Shows grid-scale cryogenic storage and waste-heat coupling

v8 answer (Section 3, row 9): 180 °C waste heat drives the expansion and the loop head; no electric pumps.

2.7 Waste-heat availability

Metric	Demonstrated (A)	Conditions	Source	v8 parameter	Ratio	Extrapolation note
Global waste heat	72 % of primary energy lost after conversion; 63 % of streams below 100 °C	Global top-down estimate	[28]	Waste heat is free fuel [V8]	n/a	Most waste heat is low-grade; v8's booked point uses 180 °C heat, so only the hotter fraction applies directly
Data-centre electricity	~415 TWh (2024); ~945 TWh projected (2030)	IEA, Energy and AI	[29]	457.66 MW of waste heat closes one module's books	≈ 47.4 GW average ≈ 104 modules' worth *(derived here)*	Data-centre heat is low-grade; v8 itself limits low-grade heat to a fraction (v7 erratum 6). Upper bound only

v8 answer (Section 3, row 10): waste heat is the fuel; the books close on the booked point's own waste heat.



Demonstrated (A) vs v8 paper values (B), log scale. Not like for like.

3. How SCG-HMH addresses the limits of the established benchmarks

Each benchmark in Section 2 was held back by documented limits, including plasma confinement, ionisation instability and recombination (rows 11–12). This section sets each limit beside the v8 mechanism that answers it. The mechanisms are taken only from the locked v8 paper [V8] and the SCG-HMH records it cites; the effects are those v8 derives. They are design answers derived on paper. They count as confirmed only when the bench and the staged tests prove them.

3.1 Limit to mechanism

#	Benchmark limit (documented)	v8 mechanism [V8]	Effect (v8 number or qualitative)	Status
1	Open-cycle MHD needs thermal ionisation: seeded combustion plasma at about 2,600 °C inlet [16], seed recovery [16], electrode damage and large near-electrode voltage drops [30], and works at about 5 S/m [17]	Passive ionisation of cold, charged cryogenic N_2 (rotor dB/dt, shear, seed heat); no thermal-ionisation (Saha) or hot-plasma model	No combustion, no alkali seed, no slag; σ_high = 14,000 S/m derived from published physics	Derived on paper; Phase-1 gate (σ > 5 S/m, > 50 W net) to confirm
2	Closed-cycle non-equilibrium MHD depends on seed fraction; seed fluctuation and wall attachment cut output [18][19]	Low-duty Marx pulse pre-ionisation (1.0 W, fixed) plus positive N_2 chamber pressure (published lever)	σ ×1.2377 at 8 bar g (17,328 S/m); the same output at lower rpm (ω ∝ σ^(−1/2))	Derived on paper; pressure step-up to confirm

#	Benchmark limit (documented)	v8 mechanism [V8]	Effect (v8 number or qualitative)	Status
3	HTS machines carry a separate cryogenic plant: the 20 T coil needed 600 W of cooling at 20 K [1]	ReBCO stators run at 77–85 K on the same LN_2 sink as the working loop; the cryo load sits inside the 0.8 kW module budget	Cryogenics integrated, not a parasitic add-on; COP_e = 124,437 on 0.8 kW	Derived on paper; stator temperature to measure (F1, S2) WHAM (2024) confined plasma with 17 T REBCO HTS mirror magnets, the first HTS magnets in a mirror fusion experiment [44]
4	Fixed-field machines: output grows only as B^2	Published back-EMF feedback raises the stator field with rpm; loop gain G = g(1 + f_rel)	Every amplifier that raises rpm earns n = 2 + 2g/(1 − g) instead of 2; g_0 = 0.10 gives 5.565 T and 126.96 MW at SF 2; ceiling 20 T	Derived on paper; g_0 to measure on the bench (S1)
5	HTS bearings tested to about 15,000 rpm, with drag and a mechanical clutch limiting the test [12]	Dual YBCO flux-pinned bearings up-rated to 1,300 N, passive, 0 W, in the 77 K loop	31.7 cm^2 needed against 260.2 cm^2 per end available	Derived on paper; spin step-up to confirm
6	Rotor side loads and whirl in shafted machines [22]	Four symmetric stators centre the rotor; hub-free rotor	Net magnetic side load zero when centred; whirl limit 276,000 rpm (published) above the booked 150,938 rpm	Derived on paper; bench whirl logging
7	Composite rims limited by hoop stress; a shafted hub with keyways doubles peak stress (K_t ≥ 2)	Torque applied as a distributed Lorentz force (K_t = 1); levitated, hub-free; Maxwell-stress relief from the ReBCO stator	Same sleeve runs ×1.4142 faster; relief up to 0.497 GPa at 5 T; 150,938 rpm continuous at SF 2, 206,712 rpm at burst	Derived on paper; staged spin-up 10k → 50k → 120k (published)
8	LN_2 engines get low specific work from low-grade heat: 174 kJ/kg with 30–60 °C fluid [26]	180 °C waste heat drives the expansion; regeneration recirculates heat inside the loop	w_exp = 242.5 kJ/kg; 298.47 MW per module recirculated	Derived on paper; expansion-loop test
9	Pumps and liquefaction cost electricity	The tank head from waste heat and the generator suction make the expansion channel one thermodynamic pump, with no electric pumps	32.95 bar available against 8.00 bar needed; its heat is 0.263 % of the waste heat	Derived on paper
10	Waste heat is low-grade and dispersed: 63 % of streams below 100 °C [28]	Waste heat is the fuel (unpaid); the module train shares one LN_2 tank with inter-module resets; the AI-oracle holds the train	Books close on 457.66 MW per module; a 100-module train gives 9.9549 GW, held to 9.9052–10.0047 GW	Derived on paper; site heat grade sets which sites qualify

#	Benchmark limit (documented)	v8 mechanism [V8]	Effect (v8 number or qualitative)	Status
11	Conventional MHD channels must contain the plasma: ionisation instability in non-equilibrium generators [31][32], and wall attachment and seed deposition [18]	The spinning EMF in the toroidal ring traps the plasma passively: the rotor is a “magnetic trap” [R3]; nuclei “experience trapping, gyromotion, and azimuthal drifts/circulation within the annular MHD channel” [R2]; carriers are advected straight to the stators, “no confinement” [R1]	Zero external confinement budget; no confinement magnets or power beyond the booked 0.8 kW module budget	Derived on paper (author-confirmed design mechanism, prior SCG-HMH records); bench to confirm. Public analogues (partial/adjacent): MCX, supersonic E×B rotation with axial density drops above 10× and stability over thousands of flute times [35][36]; CMFX, rotation above 1,700 km/s, discharges to 9.5 s [37]; both are driven electrically by a biased electrode, so their confinement costs input power. LDX, levitated superconducting dipole, self-organised inward pinch with no added confinement power [38]
12	Recombination and ionisation decay limit how long conductivity lasts, so external ionisation must be kept on	The ReBCO stators reduce the recombination rate: “ReBCO stators … reduces three-body recombination losses by >90 %” [R1]; electrons are caught by the ReBCO stators and electrodes	σ sustained with no ionisation input beyond the 1.0 W Marx; the size of the residual against recombination is listed as an open quantitative item in [R4]	Derived on paper (author-confirmed); bench to confirm. Partial support: strongly magnetised three-body recombination ≈10× lower than at B = 0 (theory and simulation) [39][40]. Contrary/complicating: measured antihydrogen formation scaled only as T^(−0.7±0.2), not the simple law [41]; an 850 G field slowed an N_2 afterglow decay by only 2.9 ± 0.5 % [42]; N_2 dissociative recombination (two-body, $\alpha = 4.0 \times 10^{-7}$ cm^3/s) is not the channel a field suppresses [43]

3.2 Why integration compounds

The levers multiply. Output follows the ring equation $P = \sigma(\omega r B)^2 K(1 - K) \cdot 2\pi r h \delta$, so a gain in σ, in ω and in B each multiplies the others: per module, r^3 (+33.10 % for +10 %), B^2 and ω^2 (+21.00 %), then σ, h and δ (+10.00 %). Pressure raises σ, which buys the same output at lower rpm; the hub-free rotor and the magnetic relief raise the allowed rpm; back-EMF feedback turns each rpm gain into rpm^n with n > 2. As v8 states, this is growth by power laws (multiplicative), not exponential growth, and it is bounded: by the sleeve, the bearings, the whirl limit and the 20 T Ic ceiling.

3.3 Efficiency as v8 frames it

Two accounts are kept side by side. The full thermodynamic account, COP_th = P / (waste heat + electricity in), is 0.2175 per module at the booked baseline (5 T, 150,938 rpm, SF 2, 99.55 MW). The electricity-only account, COP_e = P / electricity in, is 124,437 on the 0.8 kW module budget. COP_e is high because the heat is unpaid; it stays high only while the cold side runs on

waste heat. Both are paper values; the bench meters the electricity in and the waste heat so that both COPs are measured.

3.4 Passive confinement and recombination: the SCG-HMH position and the public evidence

SCG-HMH position (as booked in the authors' earlier records). The spinning EMF in the toroidal ring traps the plasma passively, so there is no external confinement budget: the rotor is a "magnetic trap" [R3], magnetised nuclei undergo "trapping, gyromotion, and azimuthal drifts/circulation within the annular MHD channel" [R2], and carriers are advected straight to the stators rather than confined against walls [R1]. The ReBCO stators reduce recombination: the stator and channel conductors "catch" electrons flung outward by the rotor, which "reduces three-body recombination losses by >90 %" [R1]. The size of the residual ionisation power against recombination at design density is listed as an open item [R4]. These are author-confirmed design mechanisms on paper; the bench confirms them.

Mechanism backing for ionisation. Nitrogen is ionised by dense alternating magnetic fields: inductively coupled N_2 plasmas are sustained by a time-varying magnetic field alone, with measured electron densities [33][34]. In SCG-HMH the rotor's dB/dt plays the same role, so ionisation is set by rpm.

The table follows the companion's format. For each benchmark it gives what limited it, how SCG-HMH's integration overcomes that limit (using only features already in the records or the v8 paper), the remaining gap, and the bench step that closes it. Sources that support, are analogous to, or run contrary to the SCG-HMH position are labelled as such.

Benchmark (evidence class)	What limited it	How SCG-HMH optimises where they don't	Gap	Bench step that closes it
LDX levitated dipole [38] (supporting, adjacent): a levitated superconductor gives a self-organised inward pinch with no added confinement power	Hot, low-pressure, fully ionised plasma; a single static dipole; no power extraction	SCG-HMH uses the same passive principle (a superconducting field with no supports or confinement drive), with the rotor's spinning field adding azimuthal circulation in the annulus [R2][R3] and the four ReBCO stators shaping the field	Not demonstrated for cold, high-pressure, weakly ionised N_2 in a toroidal annulus	Measure the σ profile and decay time against B and rpm in the annulus
MCX [35][36] and CMFX [37] (analogous, falls short): supersonic E×B rotation confines and stabilises plasma (axial density drop above 10×; discharges to 9.5 s)	The rotation is driven electrically by a biased electrode (10–20 kV in MCX; a 100 kV / 100 kW supply in CMFX), so confinement costs input power; hot hydrogen in a mirror	The rotation is supplied by the rotor itself: the spinning EMF gives the trapping with no external drive and no confinement power beyond the booked 0.8 kW module budget [R1][R3]; the four symmetric stators centre the field	Whether rotor-driven rotation gives the same shear stabilisation in cold N_2	Log σ fluctuation and stability across the staged spin-up (10k → 50k → 120k rpm)
WHAM 17 T HTS mirror [44][46] (supporting): REBCO HTS magnets confined plasma from 15 July 2024; edge "stirring" rotation raised stored energy by 50 %	Driven edge biasing, which "can also be used to drive MHD and make confinement worse"; ECH heating power	SCG-HMH uses REBCO stators at 77–85 K on the shared LN_2 sink, inside the module budget, with no ECH heating and no biasing drive	WHAM's field is a mirror for hot plasma; SCG-HMH's field supports cold N_2	Field step-up with stator temperature measured (F1, S2)
Glinsky & O'Neil 1991 [39] and Phys. Rev. A 2025 [40] (supporting, theory and simulation): strongly magnetised three-body recombination is about 10× lower than at B = 0	Applies to the weakly correlated, collisionless three-body channel only	The booked ReBCO stator field (5 T) puts cold carriers in the strongly magnetised regime, and the stators collect electrons before they recombine [R1]	Not measured in cold, high-pressure N_2, where electron–neutral collisions may break magnetisation	Measure σ decay time against B at the design pressure

Benchmark (evidence class)	What limited it	How SCG-HMH optimises where they don't	Gap	Bench step that closes it
ATHENA 2004 [41] (contrary or limiting): antihydrogen formation in a 3 T trap scaled only as T^(−0.7±0.2), not the simple field-free law	Real magnetised recombination is regime-dependent; the trap holds a static, decaying population	SCG-HMH does not rely on recombination following the simple law over a long hold: ionisation is refreshed continuously at rotor rpm, and carriers are swept to the stators, so the population does not sit and decay [R1][R2]	The net loss rate at the design point is still unmeasured [R4]	Measure the residual ionisation power against recombination at the design density (the [R4] open item)
N_2 afterglow in an 850 G field, 1955 [42] (contrary or limiting): the field slowed afterglow decay by only 2.9 ± 0.5 %	A decaying afterglow with no source; a weak field (850 G); low pressure	SCG-HMH is not an afterglow: ionisation is sustained continuously by the rotor's dB/dt at rpm (with the 1.0 W Marx pre-ionisation), in cold pressurised N_2 under the far stronger ReBCO field; σ is controlled by rpm	The field effect at 5 T in cold, pressurised N_2 is unmeasured	Spin to rpm and measure σ with the drive held, then σ decay after a step down in rpm, at each field step
N_2 dissociative recombination, 1959 [43] (contrary or limiting): $e + N_2^+ \rightarrow N + N$ at $\alpha = 4.0 \times 10^{-7}$ cm^3/s, a two-body channel a field does not suppress	Molecular ions recombine dissociatively; no published data show a field reducing this channel	SCG-HMH does not claim the field suppresses this channel. It offsets it: ionisation is refreshed at rpm, carriers are collected at the stators, and oxygen is displaced from the loop by the published PSA/membrane step (built-in anti-attachment) [V8], so no O_2 attachment adds losses. The positive N_2 chamber pressure is the published lever for σ	Stays open: the dissociative loss at design density sets the residual of [R4]	Measure σ against pressure and rpm, and the residual ionisation power, at the Phase-1 bench
ICP N_2 ionisation [33] [34] (mechanism backing): dense alternating magnetic fields ionise N_2 and sustain measured electron densities	Driven by an external RF coil at mbar pressure, with no power extraction	The rotor's dB/dt is the alternating field, so ionisation comes with the rotation that also generates the output; σ follows rpm	Magnitude at design pressure and rpm	Spin to 150,938 rpm and measure σ (Phase-1 gate: σ > 5 S/m, > 50 W net)

Reading the evidence honestly. Public data support the direction of both SCG-HMH claims: rotation and strong fields confine and stabilise plasma, and strong fields suppress three-body recombination. Every public case except LDX drives its rotation externally, and every case is a hot or low-pressure plasma. The contrary evidence (ATHENA, the 1955 afterglow, dissociative recombination) shows that recombination in real magnetised plasmas is regime-dependent, and that the two-body dissociative channel in N_2 is not suppressed by a field. SCG-HMH's answer is integration: ionisation refreshed at rpm, cold pressurised N_2 under the ReBCO field, passive trapping by the spinning EMF, carriers collected at the stators, and an oxygen-free loop. That answer is derived on paper. Whether it holds in cold, high-pressure N_2 is the bench question, and the bench steps above close it.

4. (B) Where the v8 design goes beyond the demonstrated, and the bench step that closes each gap

v8 design value [V8]	Nearest demonstrated	Gap	Bench step that closes it
σ_{high} = 14,000 S/m in charged cryogenic N_2	~100 S/m (non-equilibrium), ~5 S/m (open cycle)	×140 / ×2,800	Phase-1 gate: σ > 5 S/m and > 50 W net; then pressure step-up to measure σ against pressure
Ring power density ≈ 2.13×10^5 MW/m^3 *(derived here)*	140 MWe/m^3	~×1,500	Follows σ and B; measured output against ω^2 and B^2 at each step

v8 design value [V8]	Nearest demonstrated	Gap	Bench step that closes it
150,938 rpm on YBCO bearings	~15,000 rpm	×10	Spin step-up on the bench with bearing drag, stiffness and whirl logged
13 T to 20 T stator at 77 K	20.1 T at ~20 K	temperature	Field step-up with stator temperature measured (F1, S2)
g_0 back-EMF feedback	none published	unknown	Measure g_0 at speed (S1)
w_exp 242.5 kJ/kg with 180 °C heat	174 kJ/kg at 30–60 °C	×1.39	Expansion-loop test with metered waste heat
Passive confinement and field-reduced recombination in cold, high-pressure N_2	Driven rotation in hot plasmas; magnetised three-body theory	not demonstrated publicly	Measure σ decay time against B and rpm at the bench
σ set by rpm (rpm → ionisation)	ICP ionisation of N_2 at mbar pressure	magnitude	Spin to 150,938 rpm and measure σ
Data-centre loop and cryo-CMOS on the cold side (new in V9, derived; see SUPPLEMENTARY_Transistor_Physics_and_Data_Centres)	77 K CMOS: 27 mV/dec, 1 V supply against 4 V at 300 K [45]	not demonstrated with SCG-HMH; nothing booked	Meter hall heat temperature and flow (f_rec); after the Phase-1 gate, an LN_2-immersed CMOS board as a metered vaporiser load

5. Readiness by subsystem (authors' assessment)

TRL-style levels below are the authors' assessment, not an independent rating.

Subsystem	Demonstrated elsewhere	At v8 conditions
REBCO stator coils	TRL 6–7 (20 T coil; 36.5 MW HTS motor)	TRL 3–4 (13–20 T at 77 K)
YBCO bearings	TRL 6 (flywheel at ~15,000 rpm)	TRL 3 (150,938 rpm)
Composite rotor and sleeve	TRL 8–9 (grid flywheels)	TRL 3–4 (SF 2 on the published sleeve, untested)
MHD conversion	TRL 6 (20 MW pilot, hot plasma)	TRL 2–3 (cold N_2 ring)
Cold-plasma conductivity	TRL 4 (non-equilibrium σ ~100 S/m)	TRL 2 (14,000 S/m, untested)
LN_2 expansion and storage	TRL 7–8 (engines; grid LAES)	TRL 4 (180 °C coupling)
Waste-heat supply	TRL 9 (resource exists)	Site-dependent
Integrated SCG-HMH module	n/a	TRL 2 (theory closed on paper [V8])

6. Conclusions

Most SCG-HMH subsystems have public precedents. Superconducting coils have held 20 T on a fusion-scale magnet; a superconducting motor has run at 36.5 MW; YBCO levitation at the pressure v8 uses is measured; MHD generators have delivered tens of megawatts to a grid; pulsed discharges reach high electron densities at near-ambient temperature; composite rotors spin in grid service; and liquid-nitrogen engines turn heat into shaft work. The step that is not demonstrated anywhere is the one that matters most: a conductivity of 14,000 S/m in charged cryogenic nitrogen, with the ring power density that follows from it, together with superconducting bearings at ten times their tested speed. That is why the v8 build path starts at the bench, with σ > 5 S/m and > 50 W net, and steps pressure, speed and field only after each stage is proven.

If those steps prove out, waste heat that the world already discards would become clean, abundant power. AI compute would no longer be bound by the grid, and could give its own heat back as electricity; abundant clean energy would help to reverse climate change, make

desalination and vertical farming practical, bring power to every community and open a path to space. None of this is proven yet. The public record shows that the parts are real; the bench will show whether the whole is.

Acknowledgements

The SCG-HMH concept and its core inspiration are the work of Morgan Elliott Smart and Jasper Keiron Smart. Ara (Grok Bot AI collaborator), an AI system, is credited as co-author for helping refine, check and document it; the human authors remain responsible for the theoretical claims.

References

[V8] Smart, M. E.; Smart, J. K. SCG-HMH: A Closed Derivative Chain for a Regenerative Nitrogen Plasma-Ring Generator, version 8 (29 October 2026); new version of Zenodo 10.5281/zenodo.23291052.

1. Hartwig, Z. S. et al. The SPARC Toroidal Field Model Coil Program. IEEE Trans. Appl. Supercond. (2024). https://doi.org/10.1109/TASC.2023.3332613
2. Experimental Assessment and Model Validation of the SPARC Toroidal Field Model Coil. IEEE Trans. Appl. Supercond. (2024). https://doi.org/10.1109/TASC.2023.3332823
3. Hahn, S. et al. 45.5-tesla direct-current magnetic field generated with a high-temperature superconducting magnet. Nature 570, 496 (2019). https://doi.org/10.1038/s41586-019-1293-1
4. National MagLab. With mini magnet, National MagLab creates world-record magnetic field. https://nationalmaglab.org/news-events/news/lbc-project-world-record-magnetic-field/
5. Full Power Test of a 36.5 MW HTS Propulsion Motor. IEEE Trans. Appl. Supercond. 21 (2011). https://doi.org/10.1109/TASC.2010.2093854
6. AMSC. AMSC and Northrop Grumman Announce Successful Load Testing of 36.5 Megawatt Superconductor Ship Propulsion Motor (13 Jan 2009). https://ir.amsc.com/news-releases/news-release-details/amsc-and-northrop-grumman-announce-successful-load-testing-365
7. Airbus. ASCENDing to new heights with cryogenic superconductivity (Dec 2023). https://www.airbus.com/en/newsroom/stories/2023-12-ascending-to-new-heights-with-cryogenic-superconductivity
8. Nilsson, E. et al. Performance of the 500 kW Superconducting DC and AC Links of the ASCEND Demonstrator at Airbus. IEEE Trans. Appl. Supercond. 34 (2024). https://doi.org/10.1109/TASC.2023.3346357
9. Practical adaptation in bulk superconducting magnetic bearing applications. Appl. Phys. Lett. (1992). https://doi.org/10.1063/1.107145
10. Enhanced levitation forces with field cooled $YBa_2Cu_3O_{7-\delta}$. Appl. Phys. Lett. (1998). https://doi.org/10.1063/1.121540
11. Durrell, J. H. et al. A trapped field of 17.6 T in melt-processed, bulk Gd-Ba-Cu-O reinforced with shrink-fit steel. Supercond. Sci. Technol. 27, 082001 (2014). https://doi.org/10.1088/0953-2048/27/8/082001
12. Strasik, M. et al. Design, Fabrication, and Test of a 5-kWh/100-kW Flywheel Energy Storage Utilizing a High-Temperature Superconducting Bearing. IEEE Trans. Appl. Supercond. (2007). https://doi.org/10.1109/TASC.2007.898065
13. Mattsson, A. C. J.; Brogan, T. R. Self-Excited MHD Generators. IAEA, Electricity from MHD Vol. III (1966). https://inis.iaea.org/records/yesad-0rr35
14. The Mark V 32 MW Self Excited MHD Generator. AIAA (2012). https://doi.org/10.2514/6.2012-3175

15. Magnetohydrodynamic power generation (IAEA/UNESCO review). Natural Resources Forum (1979). https://doi.org/10.1111/j.1477-8947.1979.tb00402.x
16. IAEA. MHD Electrical Power Generation (U-25 characteristics). https://www.iaea.org/sites/default/files/20104984253.pdf
17. Recent Progress in Closed Cycle MHD Studies. IEEJ Trans. Power Energy 111 (1991). https://doi.org/10.1541/ieejpes1990.111.3_237
18. High enthalpy extraction experiments with CCMHD Fuji-1 blow-down facility. AIAA (1998). https://doi.org/10.2514/6.1998-2921
19. High-Enthalpy Extraction Demonstration With Closed-Cycle Disk MHD Generators. J. Eng. Gas Turbines Power. https://doi.org/10.1115/1.2816533
20. As [15] (noble-gas MHD power density and enthalpy extraction).
21. Time resolved electron density and temperature measurements via Thomson scattering in an atmospheric nanosecond pulsed discharge. Plasma Sources Sci. Technol. (2020). https://doi.org/10.1088/1361-6595/aba114
22. NASA. G2 Flywheel Module Design. NTRS 20060028492. https://ntrs.nasa.gov/citations/20060028492
23. NASA. G2 Flywheel Module Operated at 41,000 rpm. NTRS 20050217267. https://ntrs.nasa.gov/citations/20050217267
24. Beacon Power. Stephentown, New York. https://beaconpower.com/stephentown-new-york/
25. Zwyssig, C.; Kolar, J. W.; Round, S. D. Megaspeed Drive Systems: Pushing Beyond 1 Million r/min. IEEE/ASME Trans. Mechatronics 14 (2009). https://doi.org/10.1109/TMECH.2008.2009310
26. Sechenyh, V. et al. The Effect of Heat Exchange Fluid Composition on the Performance of a Liquid Nitrogen Engine System. Energies 14, 1474 (2021). https://doi.org/10.3390/en14051474
27. Utility Dive. Hybrid compressed liquid air storage technology could be coming to the US. https://www.utilitydive.com/news/hybrid-compressed-liquid-air-storage-technology-could-be-coming-to-the-us/505660/
28. Forman, C. et al. Estimating the global waste heat potential. Renew. Sustain. Energy Rev. 57, 1568 (2016). https://doi.org/10.1016/j.rser.2015.12.192
29. IEA. Energy and AI: Energy demand from AI (2025). https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai
30. Harada, N. et al. Results of High Interaction Experiments in Closed Cycle MHD Facilities at Tokyo Institute of Technology (1986). https://edx.netl.doe.gov/storage/f/edx/2021/05/2021-05-20T06:55:18.654Z/50f48af8-cb52-42a2-88d7-ddd8443aed60/id1161_1986_n-harada_results-of-high-interaction-experiments-in-closed-cycle-mhd-facilities-at-tokyo-institute-of-technology.pdf
31. Ionization instability in nonequilibrium MHD generators: Effect of diffusion. Energy Conversion and Management (1981). https://doi.org/10.1016/0196-8904(81)90031-5
32. Numerical simulation of nonlinear ionization nonuniformities in nonequilibrium disk MHD generator. IEEE Trans. Plasma Sci. (1997). https://doi.org/10.1109/27.557481
33. Double and triple Langmuir probes measurements in inductively coupled nitrogen plasma. Progress In Electromagnetics Research (2011). https://doi.org/10.2528/PIER10110309
34. Effect of electron energy distributions on the electron density in nitrogen inductively coupled plasmas. Plasma Sources Sci. Technol. (2022). https://doi.org/10.1088/1361-6595/ac942b
35. Steady supersonically rotating plasmas in the Maryland Centrifugal Experiment. Phys. Plasmas 12 (2005). https://doi.org/10.1063/1.1896954
36. Confinement of Plasma along Shaped Open Magnetic Fields from the Centrifugal Force of Supersonic Plasma Rotation. Phys. Rev. Lett. 105, 085003 (2010). https://doi.org/10.1103/PhysRevLett.105.085003

37. Ball, J. L. et al. Measurements of Fusion Yield on the Centrifugal Mirror Fusion Experiment (2025). https://arxiv.org/abs/2505.23047
38. Boxer, A. C. et al. Turbulent inward pinch of plasma confined by a levitated dipole magnet. Nature Physics 6, 207 (2010). https://doi.org/10.1038/nphys1510
39. Glinsky, M. E.; O'Neil, T. M. Guiding center atoms: Three-body recombination in a strongly magnetized plasma. Phys. Fluids B 3, 1279 (1991). https://doi.org/10.1063/1.859820
40. Antihydrogen chemistry. Phys. Rev. A 111, 050101 (2025). https://doi.org/10.1103/PhysRevA.111.050101
41. Amoretti, M. et al. (ATHENA). Phys. Lett. B 583, 59 (2004). https://doi.org/10.1016/j.physletb.2004.01.009
42. Nitrogen Afterglow in a Magnetic Field. J. Chem. Phys. 23, 859 (1955). https://doi.org/10.1063/1.1742135
43. Measurements of Dissociative Recombination and Diffusion in Nitrogen at Low Pressures. Phys. Rev. 113, 1 (1959). https://doi.org/10.1103/PhysRev.113.1
44. University of Wisconsin–Madison. WHAM: Wisconsin HTS Axisymmetric Mirror. https://wham.physics.wisc.edu/
45. Yokoyama, M. et al. Short-channel-effect free 0.18 μm MOSFET by temperature-dimension combination scaling theory: design and experiment. IEEE EDL 15, 202 (1994). https://doi.org/10.1109/55.286692
46. Forest, C. B. et al. First results from the Wisconsin HTS Axisymmetric Mirror (WHAM). 8th Asia-Pacific Conference on Plasma Physics (AAPPS-DPP), Malacca (2024). https://www.aappsdpp.org/DPP2024/html/3contents/pdf/5638.pdf

SCG-HMH records (Smart & Smart, prior SCG-HMH publications):

- [R1] SCG-HMH Cold plasma physics explainer, in Zenodo record 18925333 (The SCG-HMH Giga factory Concept). https://doi.org/10.5281/zenodo.18925333 — §3.3: "the SCG-HMH does not seek to confine electrons in a bounded volume … Confinement fights natural diffusion …, wasting exergy on walls/magnets"; Recombination Suppressors: ReBCO stator and channel conductors "'catch' electrons flung outwardly by the rotor's dynamics. This setup reduces three-body recombination losses by >90%".
- [R2] SCG-HMH Consolidated Master v14.4, Zenodo record 20936018. https://doi.org/10.5281/zenodo.20936018 — "Full N7+ bare nuclei (Z=7) in the strong, spinning B-fields of the rotor + 4× stators become highly magnetized … a fraction of these nuclei experience trapping, gyromotion, and azimuthal drifts/circulation within the annular MHD channel."
- [R3] SCG-HMH Cold plasma Conductivity findings, Zenodo record 19171590. https://doi.org/10.5281/zenodo.19171590 — "Multi-function rotor (turbine + plasma driver + magnetic trap)".
- [R4] SCG-HMH Probability and Mitigations, Zenodo record 21682741. https://doi.org/10.5281/zenodo.21682741 — open item: "Magnitude of residual non-recovered power required to sustain the ionization fraction against recombination at design neutral density".